1. Project Title: AI-Powered Recruitment and Intelligent Resume Screening Portal

2. Problem Statement: Recruitment is an essential process for organizations, but traditional recruitment methods often involve significant time, effort, and manual work. Recruiters receive a large number of job applications and must manually review candidates' resumes to identify suitable applicants. This process can be time-consuming, inefficient, and challenging, particularly when organizations receive hundreds or thousands of applications for a single vacancy. Traditional recruitment systems mainly provide facilities for publishing job vacancies, collecting applications, and managing candidate information. However, many existing systems lack intelligent resume analysis, automated job-role prediction, and effective resume-job matching capabilities. As a result, recruiters may struggle to identify suitable candidates efficiently, while job seekers may find it difficult to discover vacancies that match their skills and qualifications. Artificial Intelligence (AI) and Machine Learning (ML) can help address these challenges by analyzing resume information, identifying relevant skills, predicting suitable job categories, and estimating how closely a candidate's qualifications match a particular job description. To address these problems, this project proposes an AI-Powered Recruitment and Intelligent Resume Screening Portal, a centralized web-based platform that simplifies recruitment management and integrates intelligent resume screening capabilities. The system will provide separate facilities for Admins, Recruiters, and Candidates. It will use machine learning models to support preliminary resume screening, improve candidate-job matching, and help recruiters make more informed recruitment decisions. Final hiring decisions will remain under human control.

3. Objectives:

1. To develop a centralized web-based recruitment management platform.
2. To provide separate dashboards and functionalities for Admins, Recruiters, and Candidates.
3. To simplify recruiter registration, verification, and approval by Admins.
4. To enable recruiters to create, publish, update, and manage job vacancies.

5. To allow candidates to create profiles, upload resumes, search for jobs, and submit applications.
6. To develop an AI-based resume job-role prediction model that identifies suitable job categories from resume information.
7. To develop an intelligent resume-job matching model that estimates the relevance between candidate resumes and job requirements.
8. To identify matching and missing skills through skill-gap analysis.
9. To provide personalized job recommendations based on candidate skills and qualifications.
10. To improve application tracking, interview scheduling, and recruitment communication.
11. To maintain secure candidate information and recruitment records.
12. To improve recruitment efficiency while ensuring that final hiring decisions remain under recruiter control.

4. Proposed Features:

4.1 Admin Dashboard and Features

4.1.1 Admin Dashboard Overview

The Admin Dashboard provides an overview of the entire recruitment platform.

Dashboard component	**Description**
Total Recruiters	Total registered recruiters
Pending Recruiters	Recruiters waiting for approval
Approved Recruiters	Recruiters approved by Admin
Rejected Recruiters	Recruiters rejected by Admin
Total Candidates	Total registered candidates

Total Job Posts	Total jobs published
Active Jobs	Currently active vacancies
Total Applications	All submitted applications
Shortlisted Applicants	Total shortlisted candidates
Rejected Applicants	Total rejected applications
Interviews Scheduled	Total scheduled interviews
Recent Activities	Latest platform activities
ML Analytics	Model performance and screening statistics

4.1.2 Authentication and Account Management

- Admin login and logout.
- Forgot password functionality.
- Secure email password reset link.
- Change password.
- Manage admin account settings.

4.1.3 Recruiter Management

- View recruiter registrations.
- View recruiter details and company information.
- Approve or reject recruiter accounts.
- Record rejection reasons.
- Send on-site and email notifications about approval or rejection.
- View approved, pending, and rejected recruiters.

4.1.4 Candidate Management

- View registered candidates.
- View candidate profile information.
- View candidate application history.
- Monitor candidate activity where authorized.

4.1.5 Job and Application Management

- View all job postings.
- Monitor active and inactive vacancies.
- Activate or deactivate inappropriate job posts.
- Remove inappropriate or invalid job postings.
- Monitor candidate applications and recruitment activities.

4.1.6 Machine Learning Monitoring

- View resume job-role prediction statistics.
- View resume-job matching statistics.
- Monitor model accuracy and evaluation metrics.
- View screening results and application statistics.

4.1.7 Activity Log and Audit Trail

- View important user activities.
- Track admin and recruiter actions.
- Monitor account approval and rejection activities.
- Track job creation, updates, and deletion.
- Monitor security-related events.

4.2 Recruiter Dashboard and Features

4.2.1 Recruiter Dashboard Overview

Dashboard component	Description
Total Jobs	Total jobs posted by recruiter
Active Jobs	Currently active vacancies
Inactive Jobs	Deactivated vacancies

Total Applicants	Total applications received
Pending Applications	Applications waiting for review
Shortlisted Candidates	Shortlisted applicants
Rejected Candidates	Rejected applications
Interviews Scheduled	Upcoming and completed interviews
Recent Applications	Latest candidate applications
AI Screening Results	Resume prediction and matching scores
Notifications	New applications and system updates

4.2.2 Authentication and Account Management

- Recruiter registration.
- Recruiter login and logout.
- Forgot password functionality.
- Secure email password reset link.
- Change password.
- Google/LinkedIn social login where enabled.
- Display recruiter account approval status.
- Restrict job-management access until Admin approval.

4.2.3 Recruiter Profile Management

- View recruiter profile.
- Update personal information.
- Update company information.
- Upload or change profile picture.
- Update company description and contact information.

4.2.4 Job Management

- Create new job vacancies.

- View published job posts.
- Edit existing job posts.
- Delete job posts.
- Activate or deactivate job vacancies.
- Specify job title and description.
- Add required skills.
- Specify education requirements.
- Specify experience requirements.
- Add salary range.
- Specify job location.
- Set application deadline.

4.2.5 Applicant Management

- View applicants for the recruiter's own jobs.
- View candidate profile information.
- View uploaded candidate resumes.
- Download resumes where authorized.
- View predicted job role.
- View resume-job matching score.
- View matching and missing skills.
- Filter and sort candidates by relevant screening information.
- Mark applications as Under Review, Shortlisted, Rejected, Interview, or Hired.
- Perform bulk application management, such as selecting multiple applications for status updates.

4.2.6 AI-Based Resume Screening

- Analyze candidate resumes.
- View predicted job categories.
- View resume-job matching percentages.
- Identify matching skills.
- Identify missing or insufficient skills.

- Compare candidates based on screening results.

4.2.7 Interview Management

- Schedule candidate interviews.
- Set interview date and time.
- Specify interview type, such as online or offline.
- Add interview location or meeting link.
- Update or cancel interview schedules.
- View interview history and status. Notify candidates about interview updates.

4.2.8 Pre-Screening Questionnaire

- Create job-related screening questions.
- Add multiple-choice or short-answer questions.
- Assign questionnaires to candidates.
- View submitted candidate responses.
- Review responses before interview selection.

4.2.9 Communication and Notifications

- Receive notifications about new applications.
- Receive Admin approval or rejection notifications.
- Send and receive messages from candidates.
- Receive email notifications for recruitment updates.
- View read/unread notification status.

4.2.10 Activity Tracking

- Track job posting activities.
- Track application status updates.
- Record bulk application actions.
- Maintain recruiter activity history.

4.3 Candidate Dashboard and Features

4.3.1 Candidate Dashboard Overview

Dashboard component	Description
Profile Completion	Candidate profile completion percentage
Available Jobs	Currently available vacancies
Total Applications	Total submitted applications
Pending Applications	Applications under review
Shortlisted Applications	Shortlisted applications
Rejected Applications	Rejected applications
Interviews	Upcoming and completed interviews
Recommended Jobs	AI-based recommended vacancies
Resume Analysis	Predicted job role and matching results
Skill Gap Analysis	Missing skills for selected jobs
Recent Job Posts	Recently published vacancies
Notifications	Application and interview updates

4.3.2 Authentication and Account Management

- Candidate registration.
- Candidate login and logout.
- Forgot password functionality.
- Secure email password reset link.
- Change password.
- Google/LinkedIn social login.
- Secure account management.

4.3.3 Candidate Profile Management

- Create candidate profile.
- Update personal information.

- Add education details.
- Add technical and professional skills.
- Add work experience.
- Add phone number and contact information.
- Upload or change profile picture.
- Specify preferred job roles and locations.

4.3.4 Resume Management

- Upload PDF resumes.
- Upload DOCX resumes.
- View uploaded resumes.
- Replace or update resumes.
- Download resumes.
- Use resume builder to create a professional resume.
- Preview resume formatting before downloading.
- Generate and download a formatted resume.

4.3.5 Job Search and Application

- Browse available jobs.
- Search jobs by title and keywords.
- Filter jobs by required skills.
- Filter jobs by location.
- View job descriptions and requirements.
- View salary range and application deadline.
- Apply for available jobs.
- Prevent duplicate applications to the same job.
- Share job vacancies using referral/share links.

4.3.6 Application Tracking

- View submitted applications.
- Track application status.
- View Pending, Under Review, Shortlisted, Rejected, Interview, and Hired statuses.

- Receive updates when recruiters change application status.
- View application history.

4.3.7 AI-Based Resume Analysis

- View predicted job role based on resume content.
- View resume-job matching score.
- View matching skills. Identify missing skills.
- Receive skill-gap feedback.
- Get personalized job recommendations based on resume information and skills.

4.3.8 Interview and Questionnaire Management

- View upcoming interview schedules.
- View interview date, time, type, and location or meeting link.
- Receive interview notifications.
- View assigned pre-screening questionnaires.
- Submit questionnaire responses.
- Track questionnaire submission status.

4.3.9 Communication and Notifications

- Receive application submission notifications.
- Receive Under Review, Shortlisted, Rejected, Interview, and Hired notifications.
- Receive email notifications.
- Send and receive messages from relevant recruiters.
- View read/unread notifications.

4.3.10 Referral and Sharing

- Share job vacancies through referral links.
- Generate candidate referral links.
- Track referral activity where supported.

4.4 AI-Based Resume Screening and Intelligent Matching

- **Model 1** – **Resume Job-Role Prediction:** Extract and process resume text, then predict suitable job categories using a machine-learning classification model.
    - **Input:** Extracted resume text.
    - **Data:** Labeled resume dataset.
    - **Processing:** Text cleaning, preprocessing, feature extraction, and classification.
    - **Training:** Train a machine-learning classification model.
    - **Output:** Predicted job category.
    - **Example** categories include Python Developer, Web Developer, Data Analyst, and Software Engineer.
- **Model 2** – **Intelligent Resume-Job Matching:** Analyze candidate resumes and job descriptions, compare candidate skills with job requirements, generate relevance or matching scores, identify matching and missing skills, and categorize candidates based on matching results.
    - **Input:** Candidate resume text, job description, and required skills.
    - **Data:** Labeled resume-job pairs or an appropriately designed evaluation dataset.
    - **Processing:** Text preprocessing, feature extraction, and similarity or classification-based matching.
    - **Training:** Train and evaluate the matching model.
    - **Output:** Resume-job relevance score or matching category.
    - **Example:**

Matching information	Example result
Resume-job match score	85%
Matching skills	Python, Django, SQL
Missing skills	Docker, AWS
Match category	High Match

4.5 Smart Job Recommendation and Skill Gap Analysis

- Recommend relevant job vacancies based on candidate skills and qualifications.

- Identify missing skills compared with job requirements.
- Provide skill-gap information to help candidates understand areas for improvement.

4.6 Additional Advanced Features

- Resume Builder: Create, preview, and download professional resumes.
- Pre-Screening Questionnaire: Allow recruiters to create job-related questions and candidates to submit answers.
- Job Sharing and Referral: Generate and share job vacancy links.
- Social Authentication: Support Google and LinkedIn login.
- Activity Log: Maintain records of important user and recruitment activities.

5. Technology Stack

The project will use the following technologies and development tools:

Category	Technology	Purpose
Frontend	React.js	Interactive user interface and dashboards
Backend	Python, Django	Server-side development and business logic
API Development	Django REST Framework	Communication between frontend and backend
Database	SQLite or PostgreSQL	Store user, job, application, and recruitment data
Programming Language	Python	Backend development and machine learning
Machine Learning	Scikit-learn	Resume classification and intelligent matching
Data Processing	Pandas, NumPy	Dataset preparation and processing
Resume Processing	PyMuPDF, python-docx	Extract text from PDF and DOCX resumes

Model Storage	Joblib	Save and load trained ML models
Styling	CSS3, Bootstrap	Responsive interface design
Frontend Programming	JavaScript	Frontend interactions and functionality
API Communication	Axios	Connect React frontend with Django backend
Email Integration	Django Email / SMTP	Password reset and email notifications
Version Control	Git and GitHub	Source code management

AI Integration

Python and Scikit-learn will be used to develop and integrate two machine-learning models:

13. Resume Job-Role Prediction Model.
14. Resume-Job Relevance Matching Model.

These models will analyze resume information and job requirements to generate predictions, matching scores, and skill-gap information.

Development and Deployment

The project will initially be developed and tested in a local development environment. Suitable cloud deployment platforms may be selected during the deployment phase.

6. Expected Outcome

The expected outcome of this project is a complete AI-powered recruitment and intelligent resume screening portal that provides a centralized platform for managing recruitment activities.

- **For Admins:** A centralized platform to manage users, approve recruiters, monitor recruitment activities, and oversee the overall system.
- **For Recruiters:** An efficient recruitment management platform to publish job vacancies, review applications, analyze resumes, identify suitable candidates, and manage interviews.

- **For Candidates:** An accessible platform to create professional profiles, upload resumes, search for suitable jobs, submit applications, receive recommendations, and track application progress.
- **For the Recruitment Process:** Reduced manual effort, better organization of applications, and improved recruitment workflow management.
- **Through AI Integration:** Automated preliminary resume analysis, job-role prediction, intelligent resume job matching, and skill-gap identification to support recruitment decisions.

The two machine-learning models will help recruiters identify potentially suitable candidates more efficiently by providing relevant screening information. Candidates will also benefit from personalized job recommendations and information about their skills in relation to job requirements.

Ultimately, the project aims to make recruitment more organized, efficient, and intelligent through the integration of web technologies and artificial intelligence, while keeping final recruitment and hiring decisions under human control.